

Basic Principles of Medicine 2

Module: Endocrine & Reproduction

Lecture:

ER 05_ Steroid Hormone Metabolism

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Acknowledgements: [Dr. Shellon Thomas](#), MD, Dr. Margit Trotz, Ph.D.

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Objectives: Steroid Hormone Metabolism

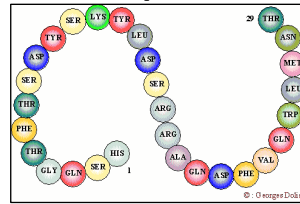
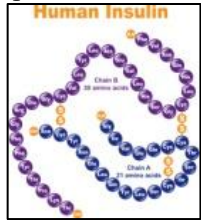
SOM.MK.II.BPM2.1.ER.2.BCHM.1024	Review the classes of steroid hormones (glucocorticoids, mineralocorticoids, androgens, estrogens and progestins).
SOM.MK.II.BPM2.1.ER.2.BCHM.1025	Indicate the steroid hormones synthesized by the different adrenal cortex layers (zona fasciculata, z. glomerulosa and z. reticularis) and the role of ACTH in their release.
SOM.MK.II.BPM2.1.ER.2.BCHM.1026	Review the synthesis and the metabolic actions of aldosterone. Indicate the role of angiotensin.
SOM.MK.II.BPM2.1.ER.2.BCHM.1027	Review the synthesis and the metabolic actions of cortisol.
SOM.MK.II.BPM2.1.ER.2.BCHM.1028	Indicate the synthesis of DHEA and androstenedione in adrenal cortex. Describe the synthesis of testosterone and estrogen from DHEA and androstenedione in peripheral tissues.
SOM.MK.II.BPM2.1.ER.2.BCHM.1029	Discuss steroid hormone synthesis related to StAR and desmolase (CYP11A). Outline steroid synthesis in the adrenal cortex.
SOM.MK.II.BPM2.1.ER.2.BCHM.1030	Differentiate congenital adrenal hyperplasias (CAH) in patients with deficiencies of 3-hydroxysteroid DH, CYP 21, CYP 11B and CYP 17 based on clinical features and laboratory findings.
SOM.MK.II.BPM2.1.ER.2.BCHM.1031	Describe the synthesis of testosterone in testes and of estradiol in ovaries. Discuss the functions of 5-alpha reductase and aromatase (CYP 19)
SOM.MK.II.BPM2.1.ER.2.BCHM.1032	Indicate the hormone abnormality in: Cushing syndrome and Addison disease.
SOM.MK.II.BPM2.1.ER.2.BCHM.1033	Indicate the biochemical basis for the occurrence of hyperglycemia in Cushing syndrome.

Recommended Reading

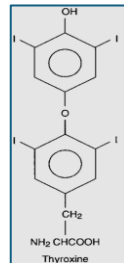
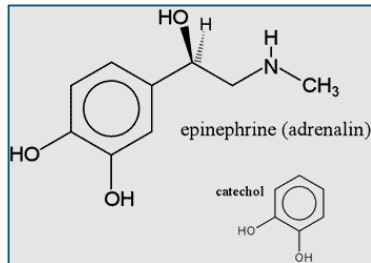
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition
Chapter 18: Pages 262 -266 [Cholesterol, Lipoprotein, and Steroid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
-  <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Mark's Basic Medical Biochemistry: A clinical approach 6th edition;
Chapter 32: [Cholesterol Absorption, Synthesis, Metabolism, and Fate | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)

Structural Classification of hormones

- **Peptide hormones:** (Examples: insulin and glucagon)

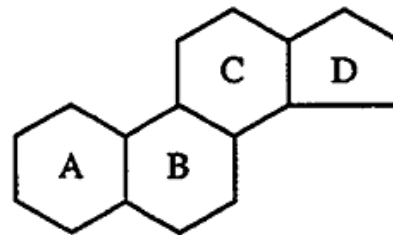


- **Amino acid derived:** (Catecholamine hormones, Thyroid hormones (iodine))



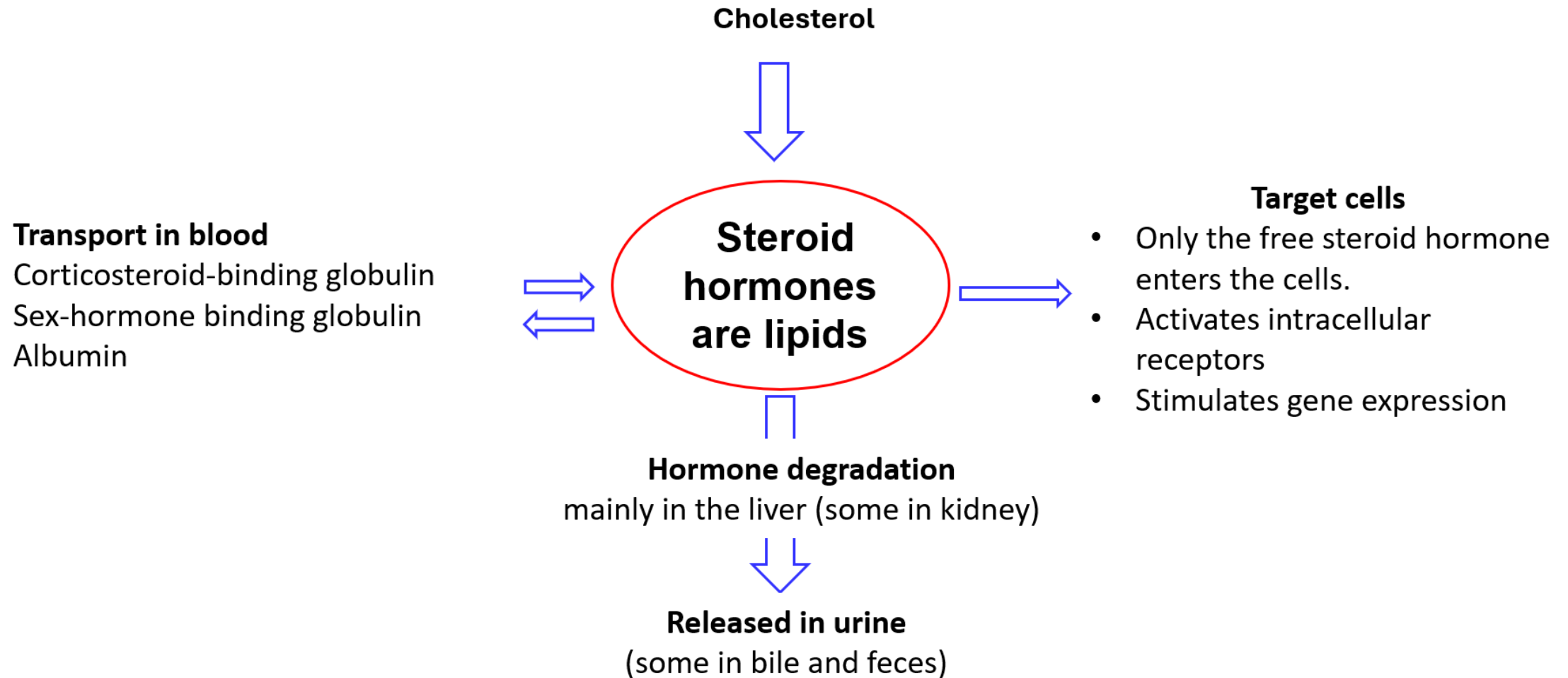
- **Steroid hormones:** (lipids, steroid ring system, 5 classes)

1. Glucocorticoids
2. Mineralocorticoids
3. Androgens
4. Estrogens
5. Progestins

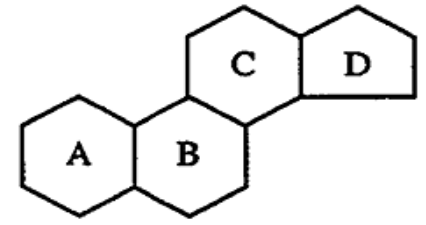


Thyroid hormones function as steroid hormones with intracellular receptors and modulate gene expression

De-novo synthesis of Steroid hormones



Functions of steroid hormones



1. Glucocorticoids C21

- **glucose** metabolism (gluconeogenesis)

2. Mineralocorticoids C21

- **mineral** metabolism
- renal reabsorption of sodium
- excretion of potassium

3. Androgens C19 (male sex hormones)

- testosterone promotes development of **male secondary sex characteristics**
- stimulates spermatogenesis, weak adrenal androgens

4. Estrogens C18 (female sex hormones)

- development of **female secondary sex characteristics**
- control the menstrual cycle

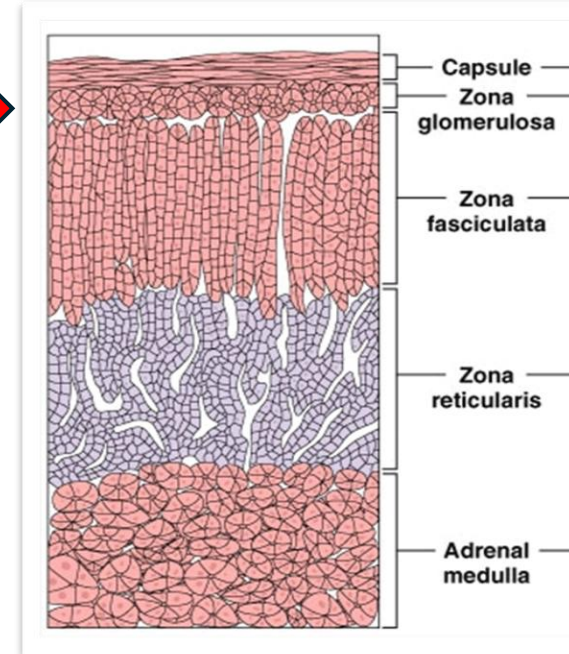
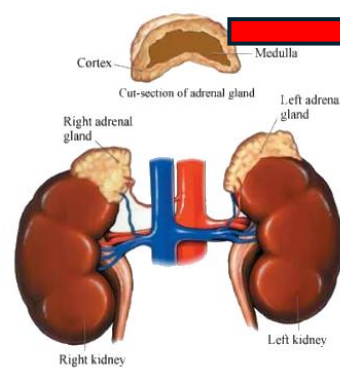
5. Progestins C21

- menstrual cycle
- pregnancy

Locations of de-novo steroid hormones synthesis

1. Adrenal Cortex:

- Cortisol
- Aldosterone
- Weak Androgens



Z. glomerulosa: **Aldosterone**

Receptors for ACTH and for angiotensin II.

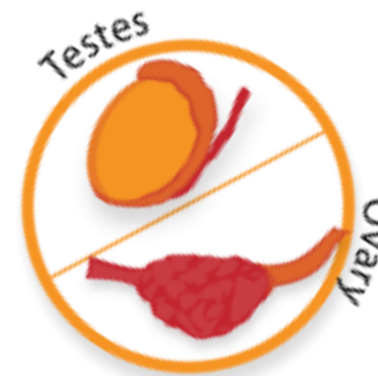
Z. fasciculata: **Cortisol** and some **DHEA**

Z. reticularis: **DHEA** and **Androstenedione**

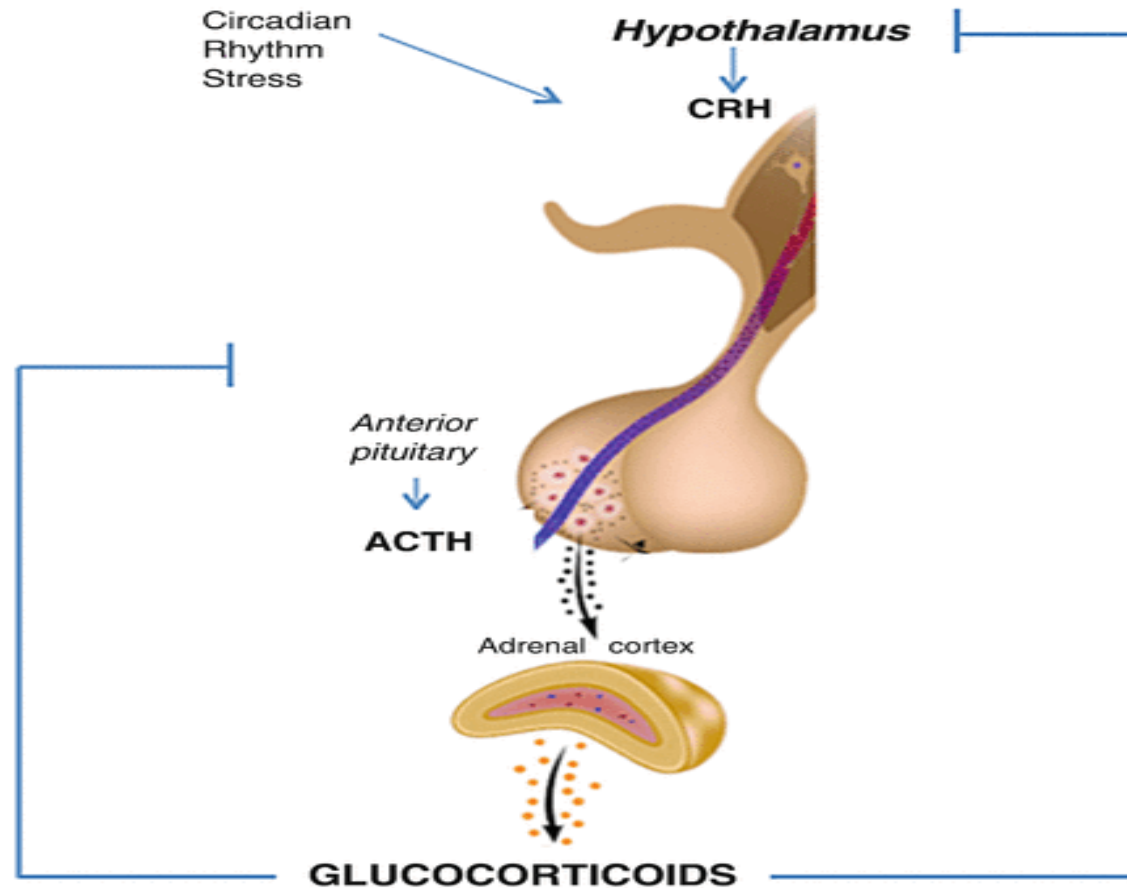
2. Testes: Testosterone

3. Ovaries: Estradiol

4. Corpus luteum: Progesterone



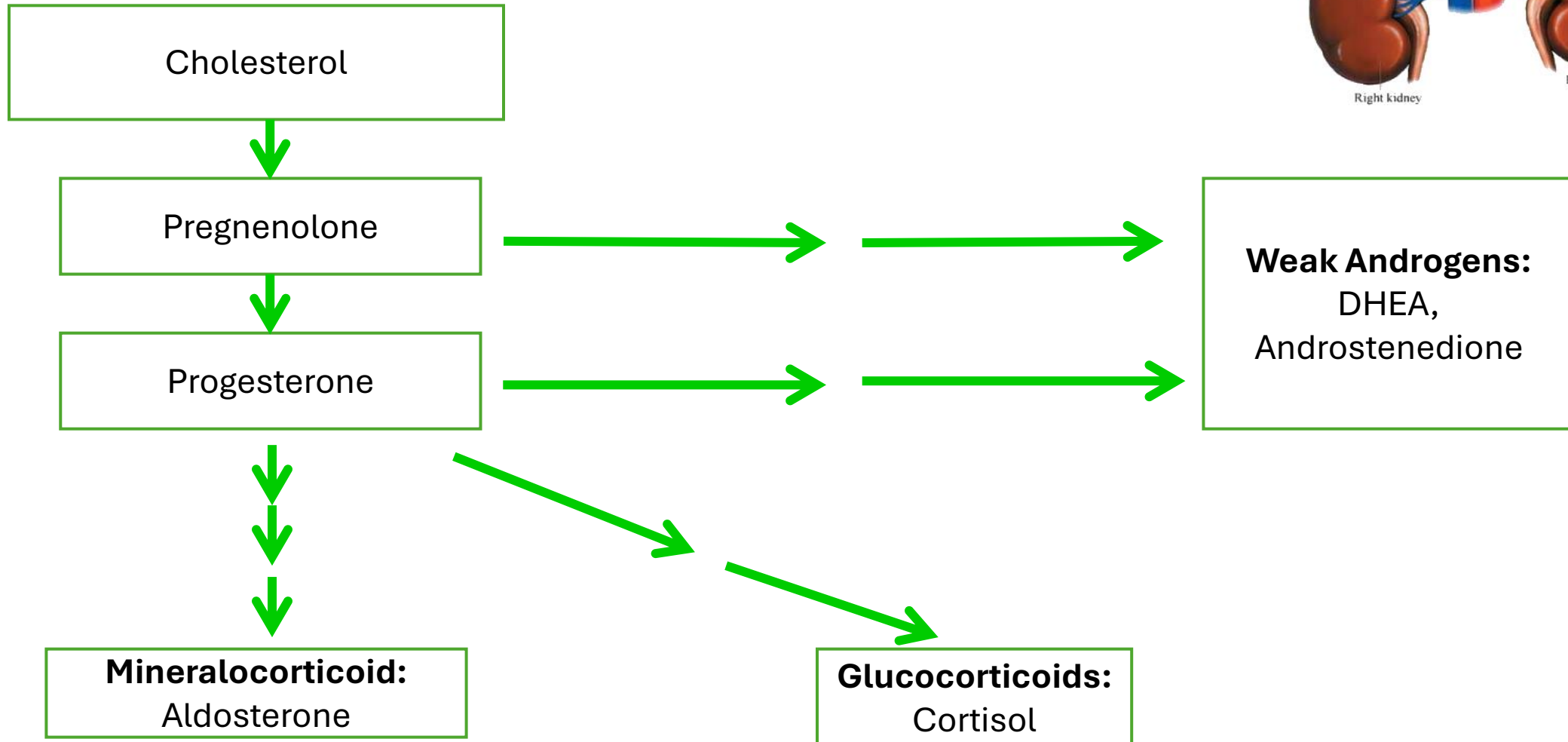
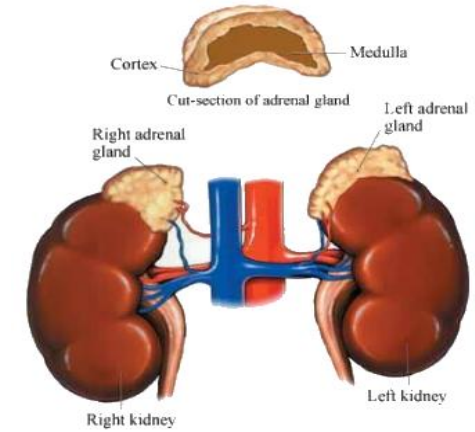
Hormonal control of the synthesis of adrenal corticosteroids



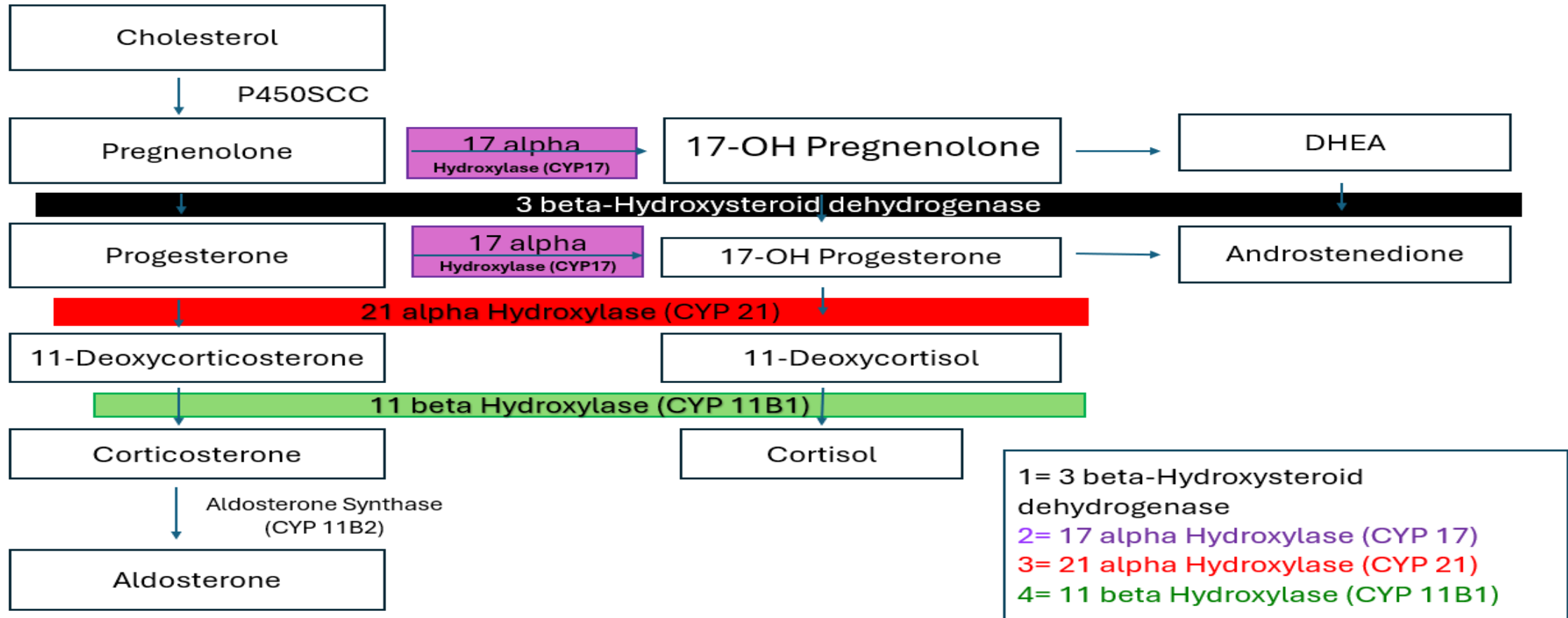
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Note: Aldosterone secretion is regulated mainly by [angiotensin II](#).

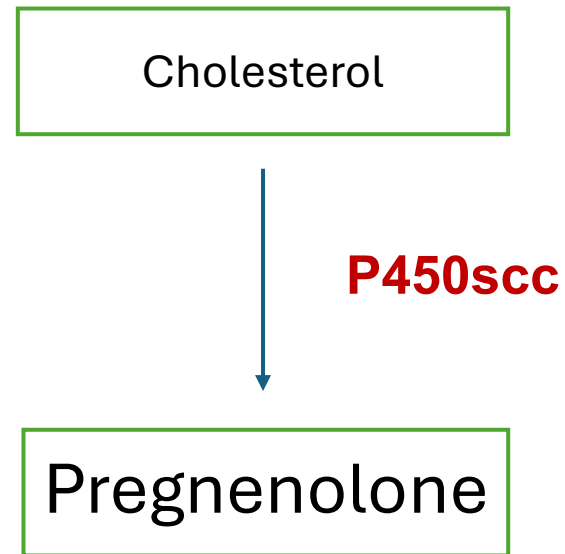
BIG Picture: Steroid hormone synthesis in the adrenal cortex



Pathway: De-novo synthesis of steroid hormones (adrenal cortex)

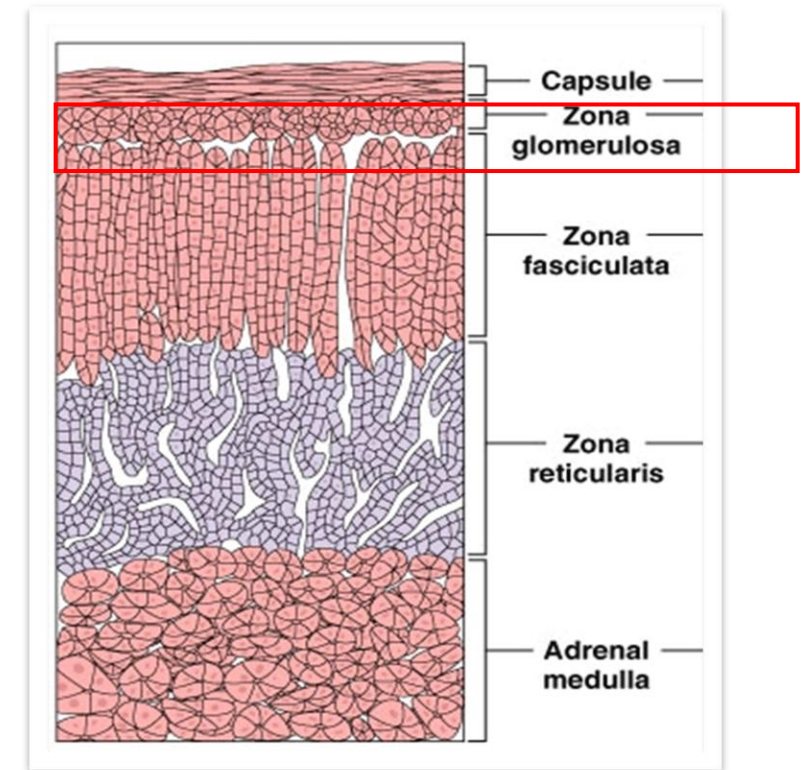
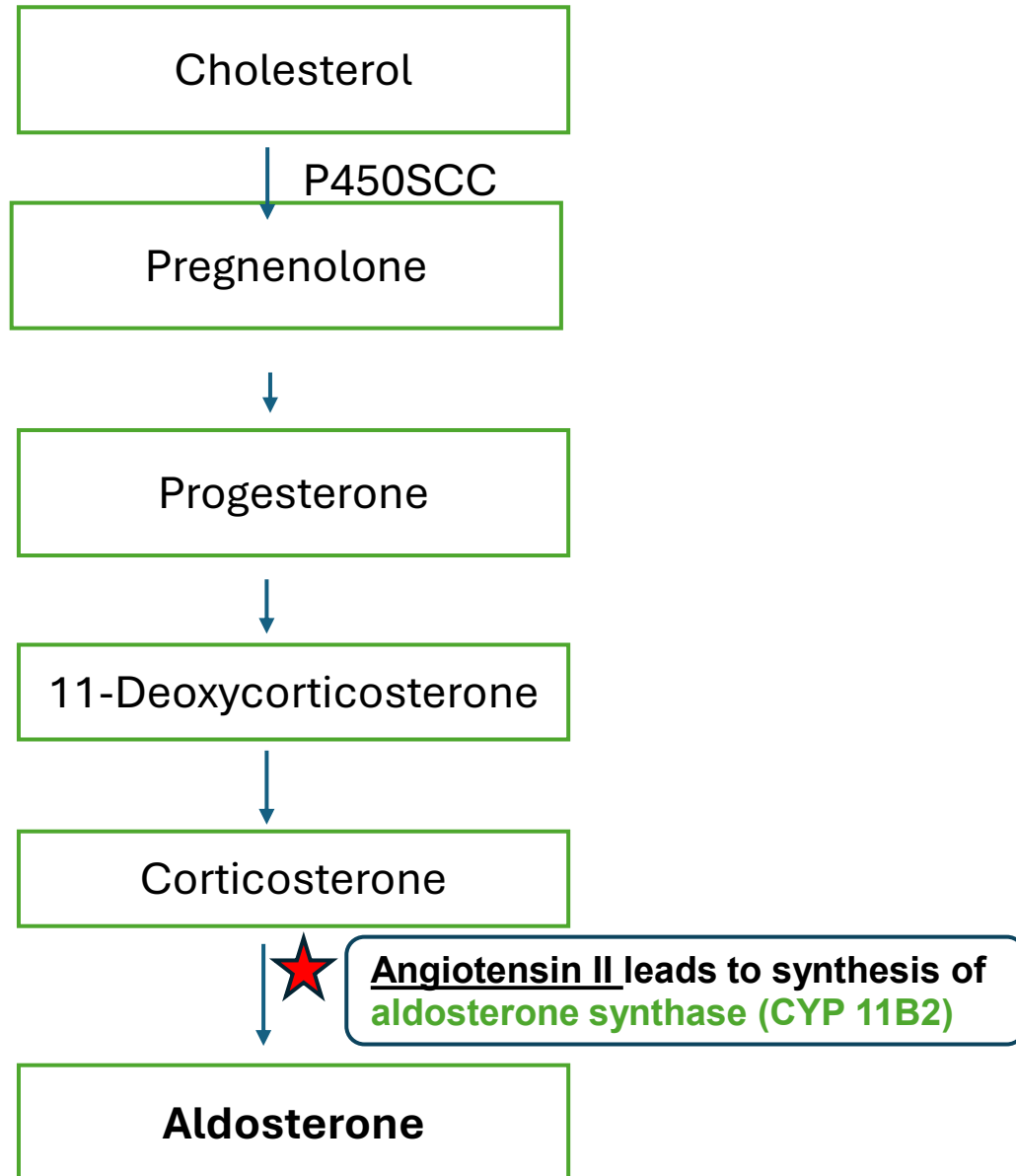


De-novo synthesis: Initial step

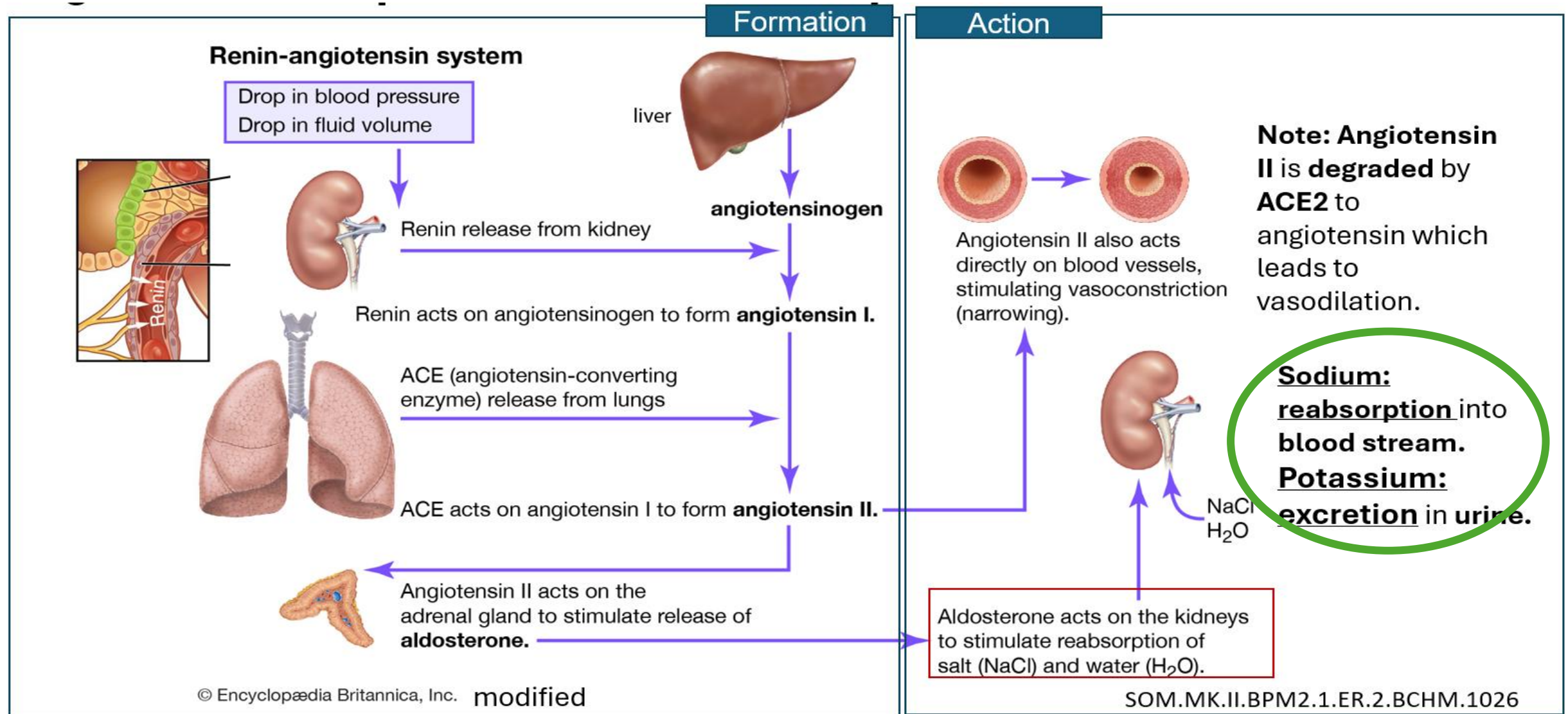


- P450scc: cytochrome P450 enzyme
- Other names: **cholesterol side-chain cleavage enzyme, desmolase or CYP 11A.**

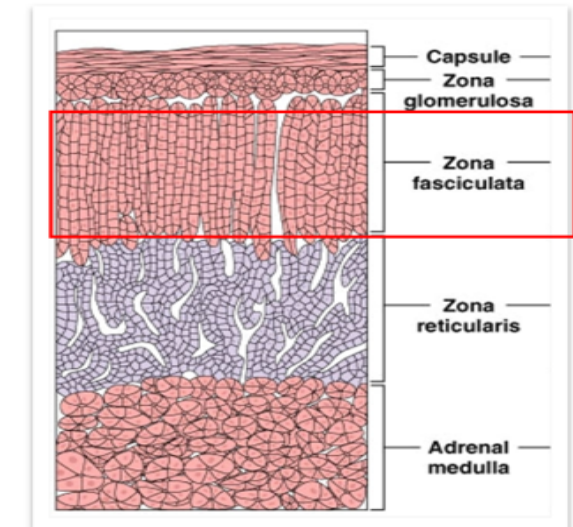
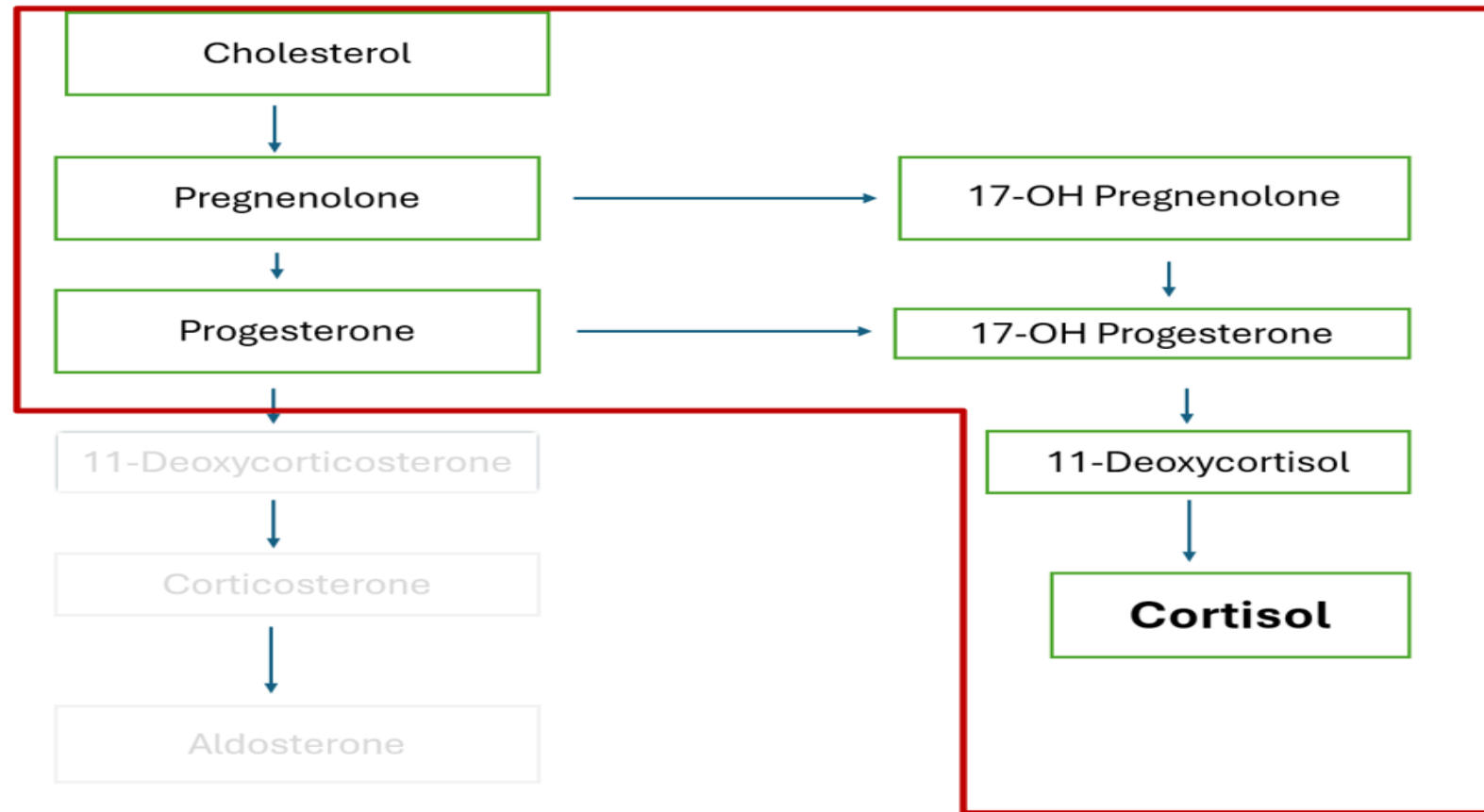
Mineralocorticoid: Aldosterone synthesis



Renin-angiotensin-aldosterone system (RAAS)



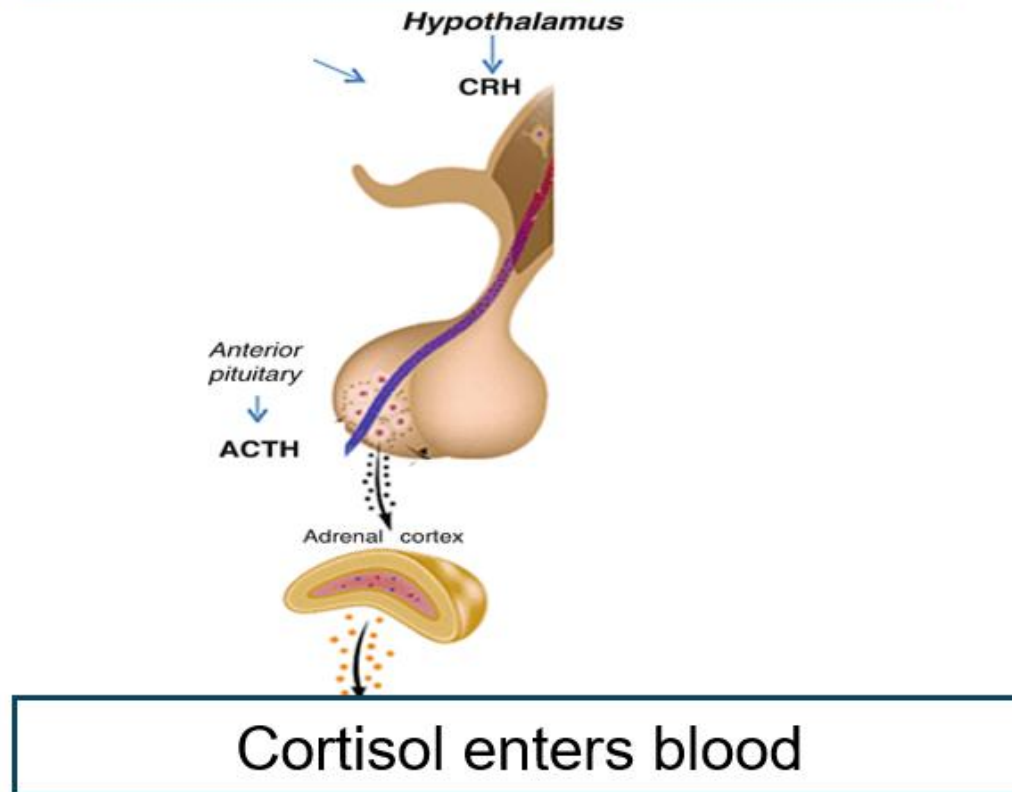
Glucocorticoid: Cortisol synthesis



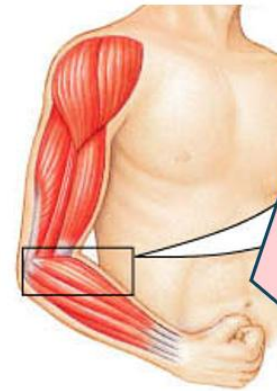
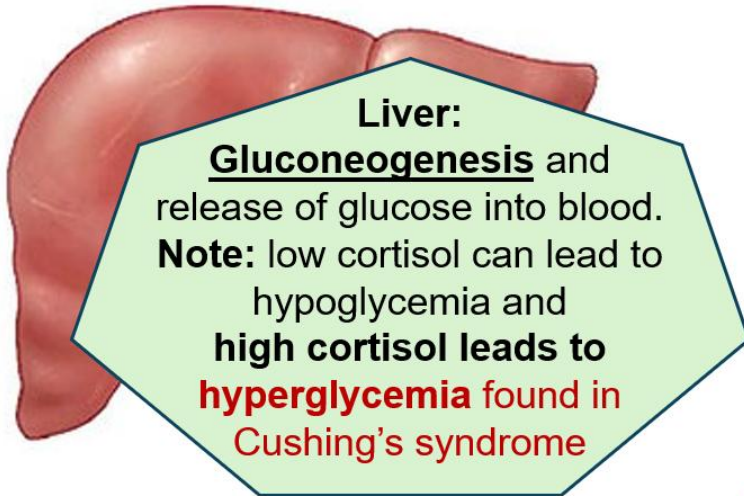
Regulation of cortisol synthesis

- Stress and low blood glucose lead to release of CRH and ACTH.

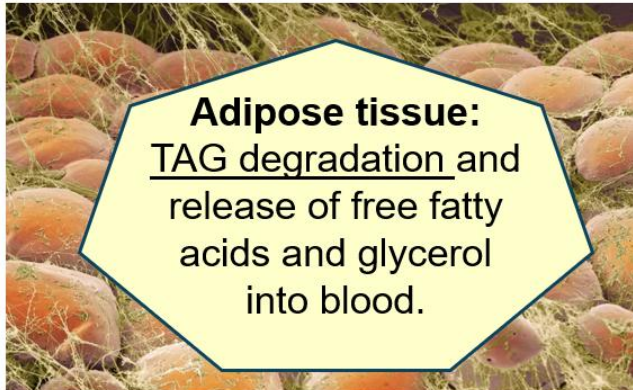
glucose-sensing neurons



Metabolic functions of cortisol



Skeletal muscle:
Protein degradation and release of amino acids into blood.

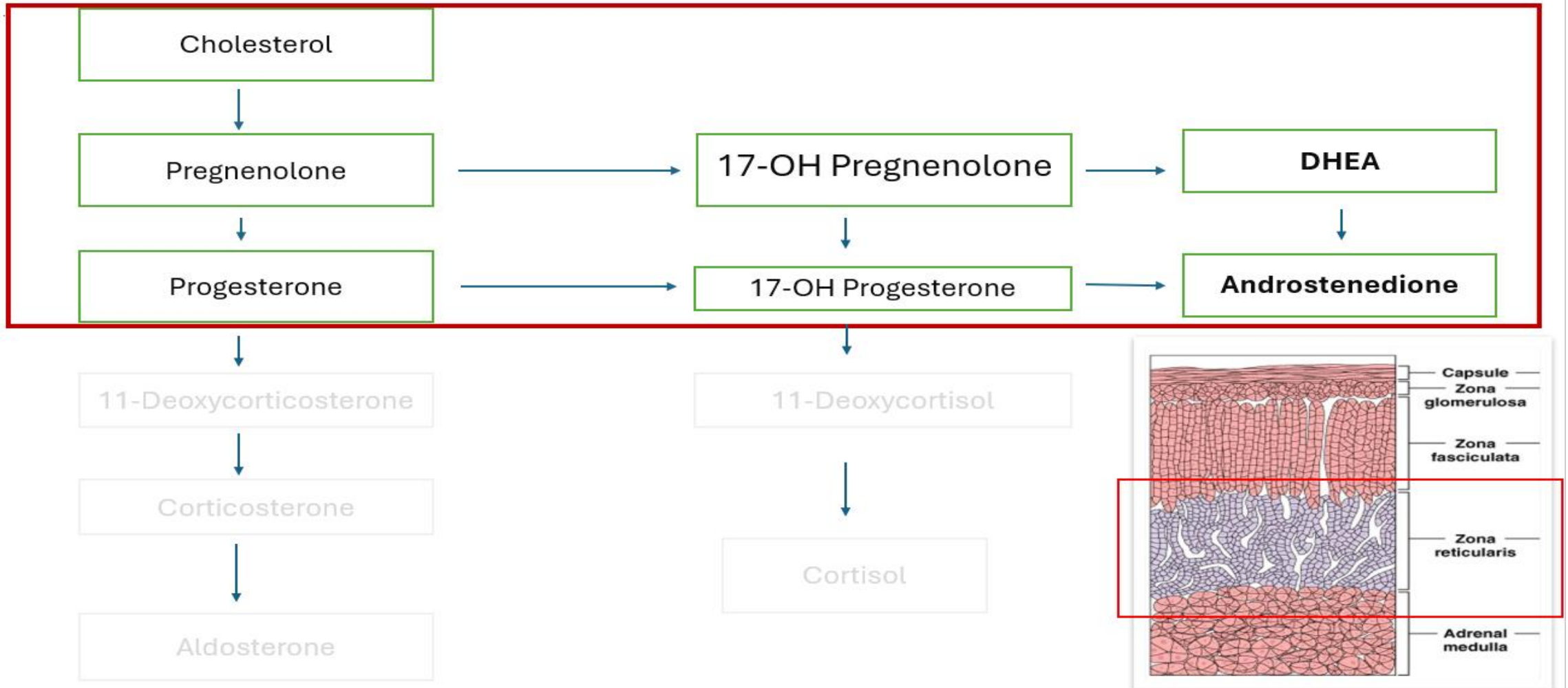


Adipose tissue:
TAG degradation and release of free fatty acids and glycerol into blood.

CORTISOL

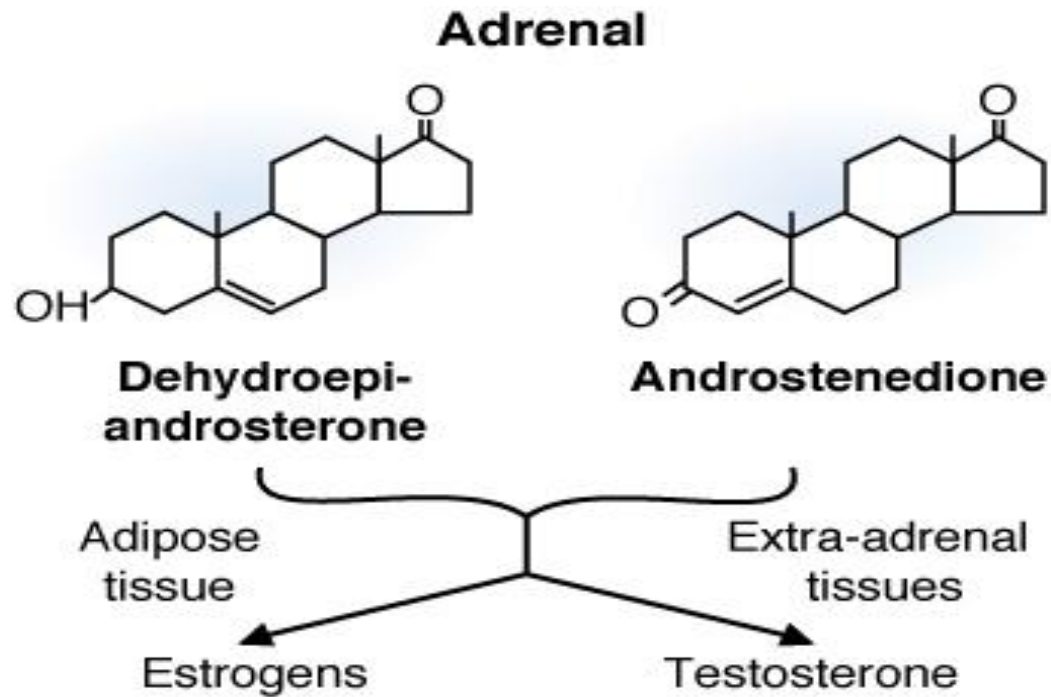
Anti-inflammatory:
Pharmacologic effect when dose exceeds physiologic range.
Inhibition of phospholipase A₂ and reduced synthesis of eicosanoids.

Sex hormones: Androgen synthesis



Fates of adrenal androgens

- Modified to estrogens or testosterone in extra adrenal tissues.

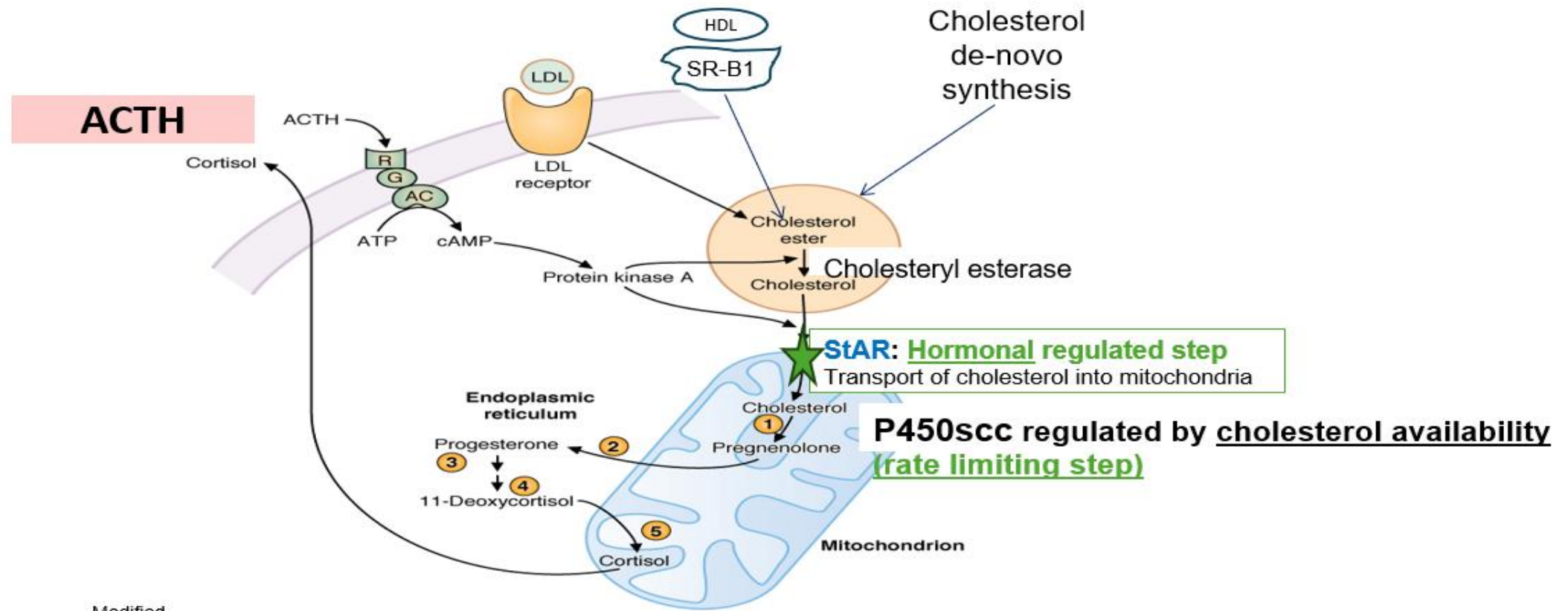


Needs **CYP 19**
Aromatase



Needs **17 β -**
Hydroxysteroid
DH-3

Regulation of *De-novo* steroid hormone synthesis

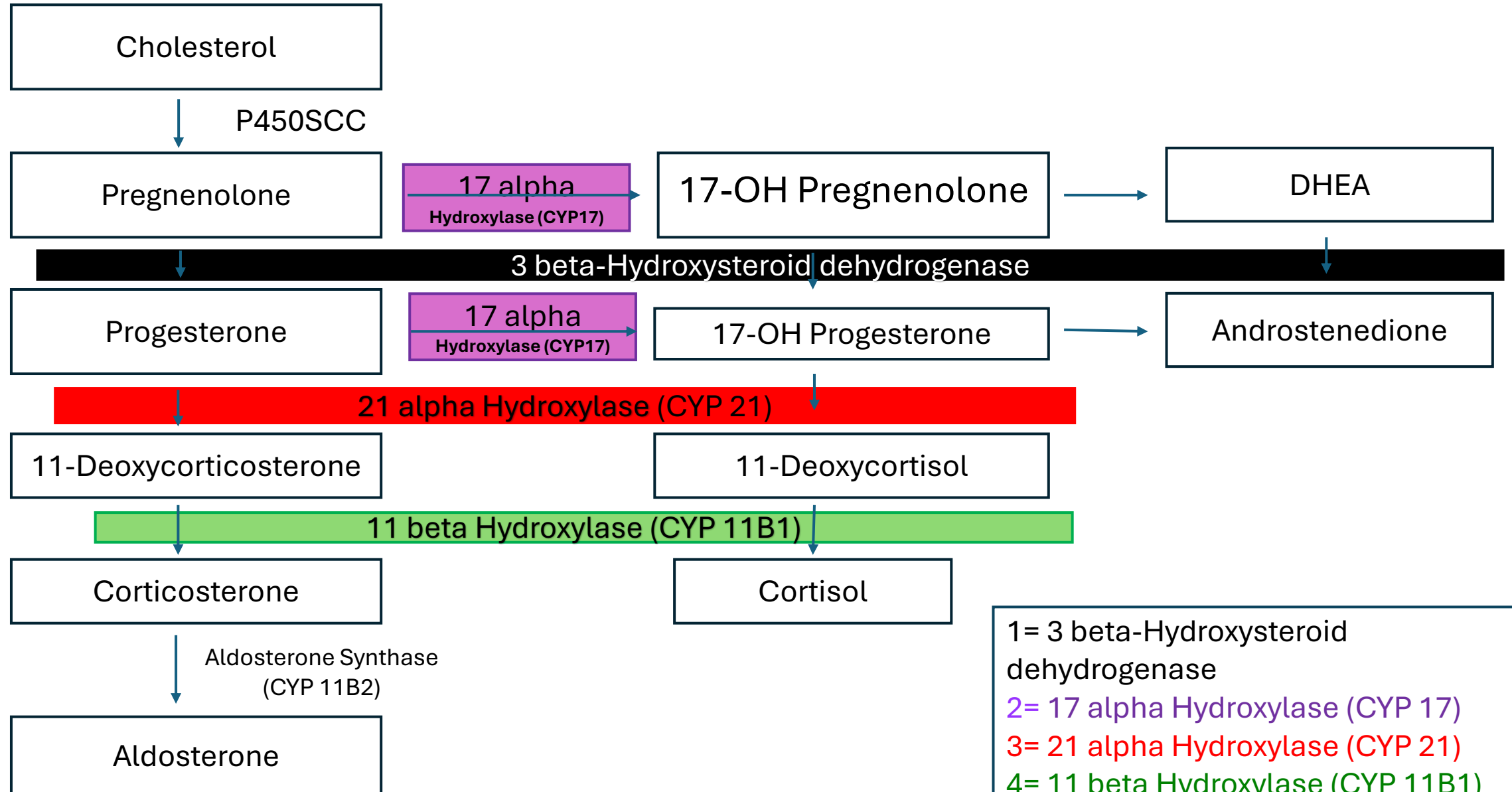


Modified
MBMB p. 654

Text: Regulation of steroid hormone synthesis in the adrenal cortex

- **Cholesterol** enters cells and is stored as cholesteryl esters in cytosol.
- **ACTH** activates ACTH receptors $\rightarrow \uparrow \text{cAMP} \rightarrow$ activated protein kinase A
- **Protein kinase A (PKA)** activates:
 - **StAR**
 - gene expression of **steroidogenic enzymes**.
- Cholesteryl esterase generates free cholesterol
- **StAR** (steroidogenic acute regulatory protein) transports free cholesterol into the **mitochondrial matrix** (**hormonal regulatory step**)
- **P450_{scc}** cleaves cholesterol to pregnenolone (mitochondria: **rate-limiting step regulated by availability of cholesterol**)

Enzymes of steroid hormone metabolism



Congenital Adrenal Hyperplasia (CAH)



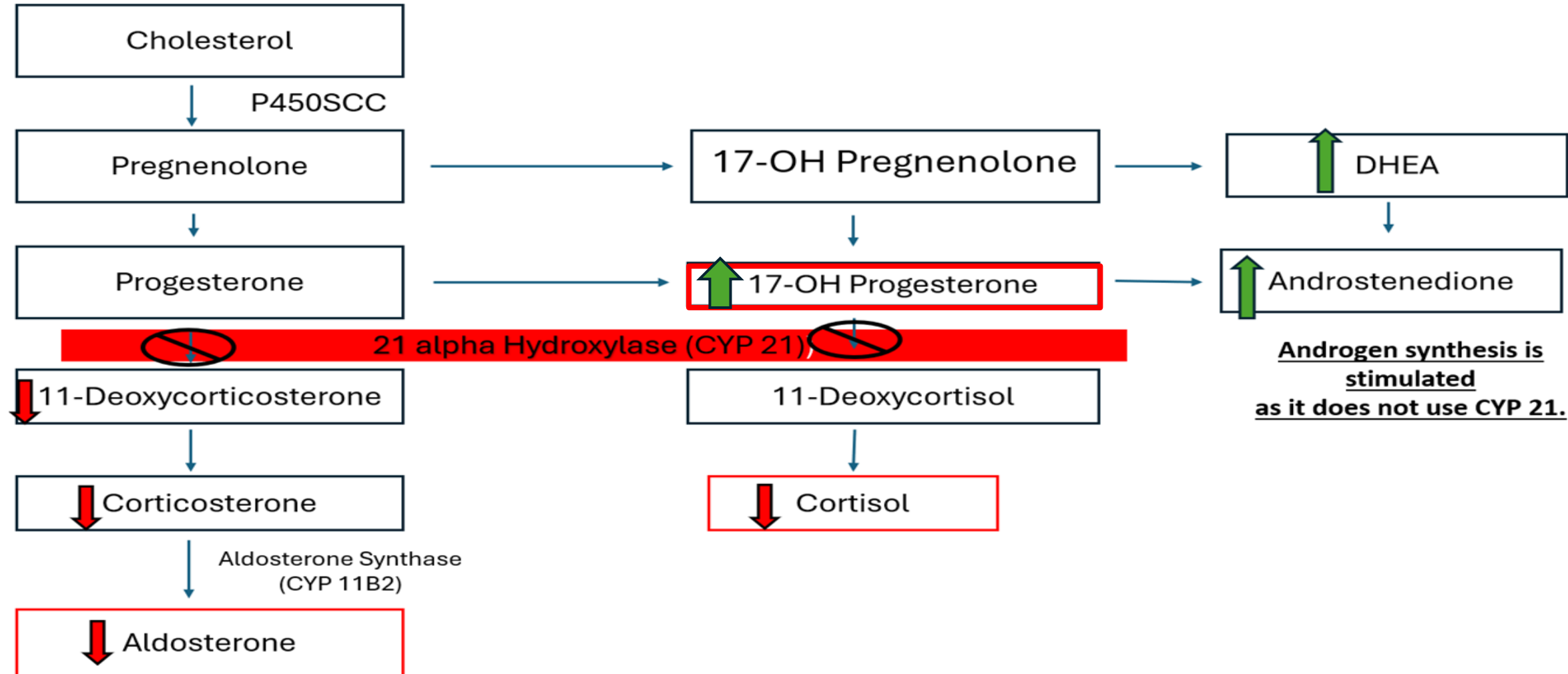
- CAH is a group of inherited (**autosomal recessive**) disorders
- One enzyme of cortisol synthesis is deficient
- Clinical spectrum of disease (most severe to mild forms)
- Depends on the degree of enzyme deficiency
- **Different** onset of clinical **manifestations**
- Low cortisol → ↓ Feedback inhibition of ACTH → Enlarged adrenal cortex

If deficient enzyme starts with 1, it causes **hypertension**;
if deficient enzyme ends with 1, it causes **virilization** in females.

Enzyme Deficiency	Blood pressure	Blood glucose	Androgens
3-Hydroxysteroid DH (rare)	Low	Low	Variable in general
17 α -Hydroxylase (CYP 17)	High	Can be low	Low in general
21 α -Hydroxylase (CYP 21)	Low	Can be low	High in adrenals
11 β -Hydroxylase (CYP 11B1)	Moderate elevation	Can be low	High in adrenals

CAH: Deficiency of CYP 21 (21-Hydroxylase)

Most common



21-hydroxylase deficiency (CYP21)

1. Classic form

a) **Low aldosterone:** (Elevated renin)

- Hypotension
- Hyponatremia (“salt wasting” crisis if severe)
- Hyperkalemia (fatal if untreated)

b) **Low cortisol:** low blood glucose

c) **Increased adrenal androgens (DHEA and Androstenedione):**

- **Female newborn:** Ambiguous male-like genitalia e.g clitoral enlargement, labial fusion. Children: Precocious puberty
- Male newborn: external male genitalia

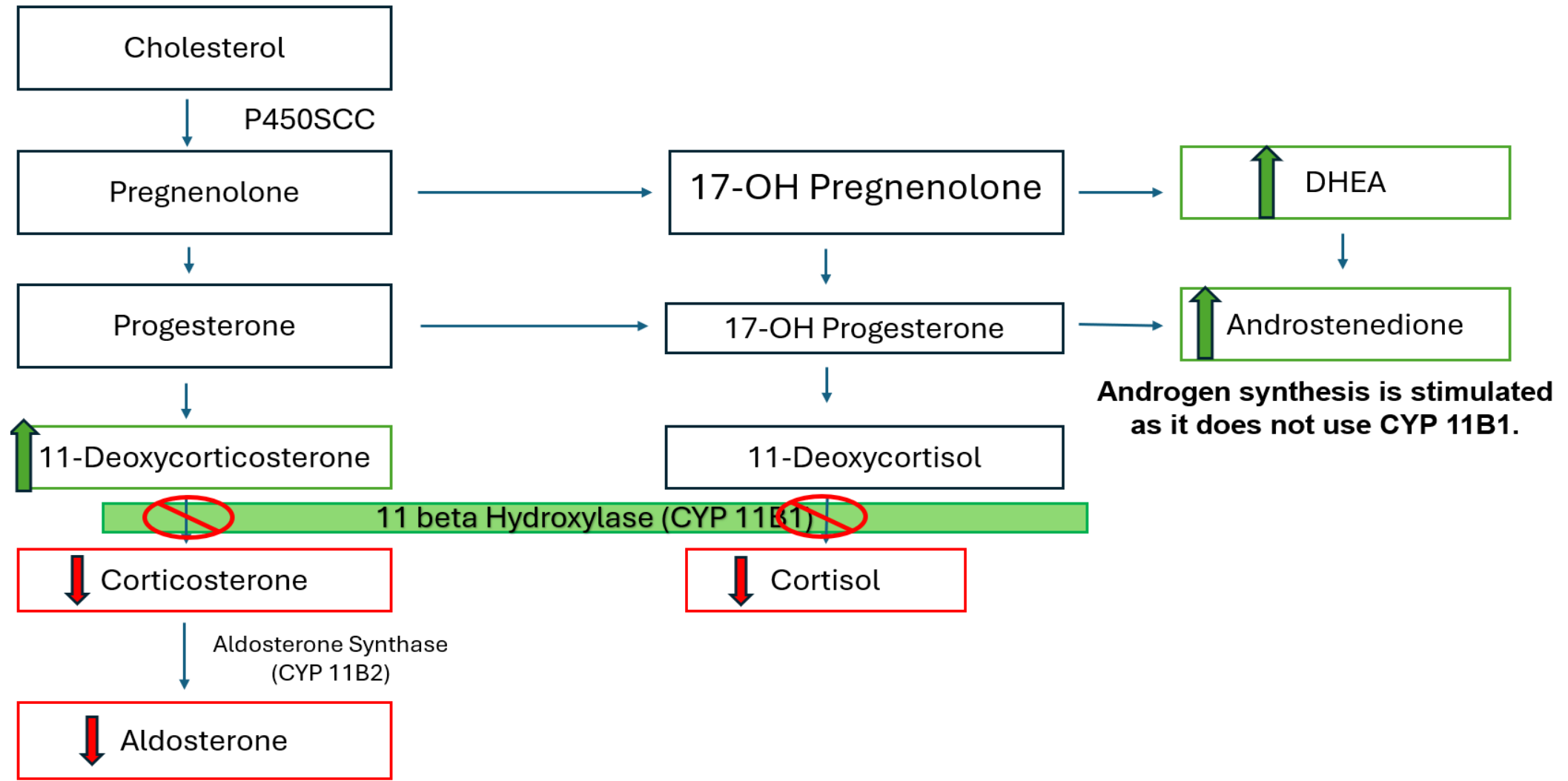


Images: UpToDate.

2. Non-classic form: asymptomatic or virilization- early pubarche, hirsutism, menstrual irregularity

- Newborn screening test for 17-OH Progesterone: **POSITIVE**

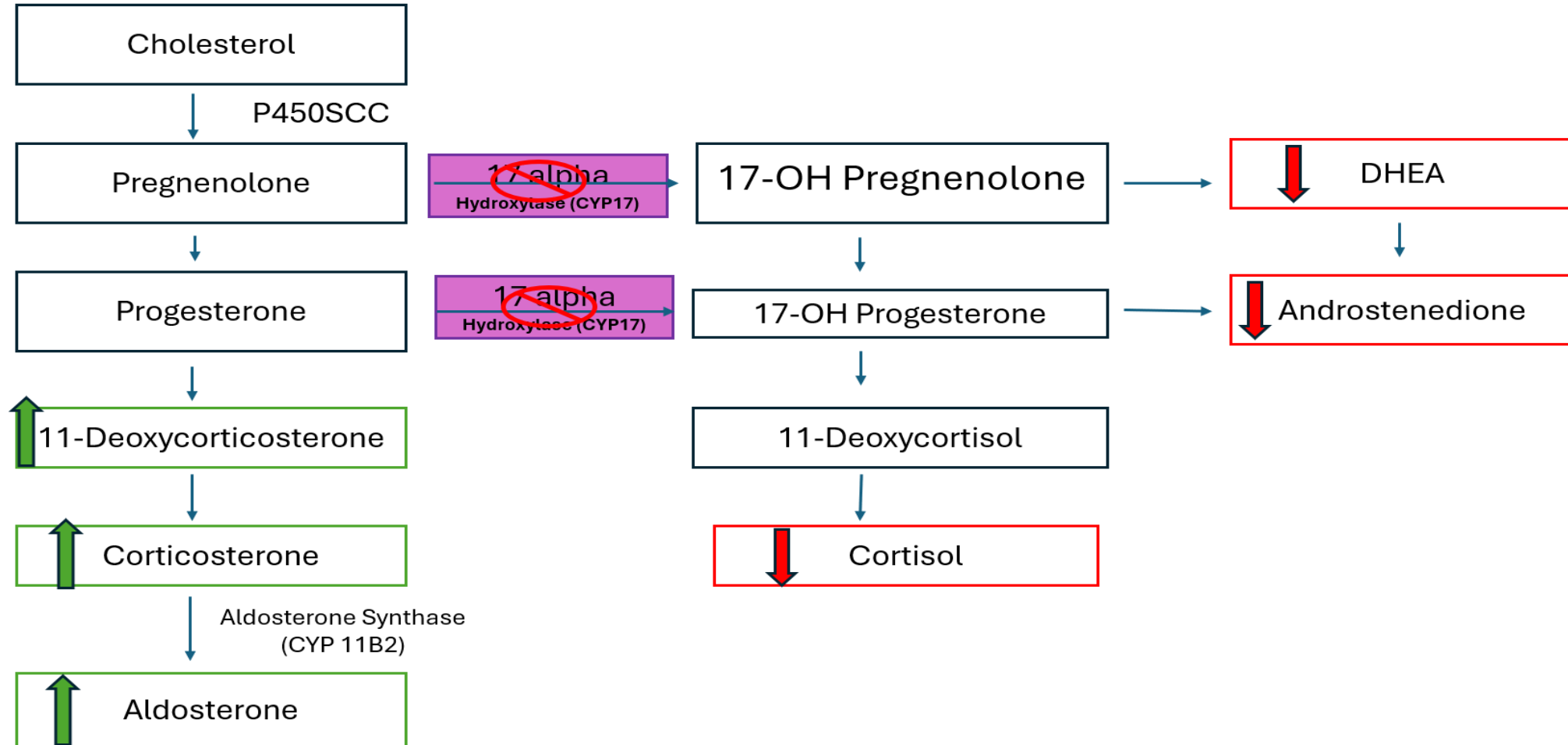
CAH: Deficiency of CYP 11B1 (11 β -Hydroxylase)



11-beta hydroxylase deficiency (CYP11B1)

- **Low aldosterone**
- **Increased 11-Deoxycorticosterone**: mineralocorticoid/aldosterone like effect
 - Moderate hypertension
 - High/normal serum sodium
 - Low/normal serum potassium
- **Low cortisol**: low blood glucose
- **Increased adrenal androgens (DHEA and Androstenedione)**:
 - **Female newborn**: Ambiguous male-like genitalia external genitalia
 - Male newborn: external male genitalia

CAH: Deficiency of CYP 17 (17α-Hydroxylase)



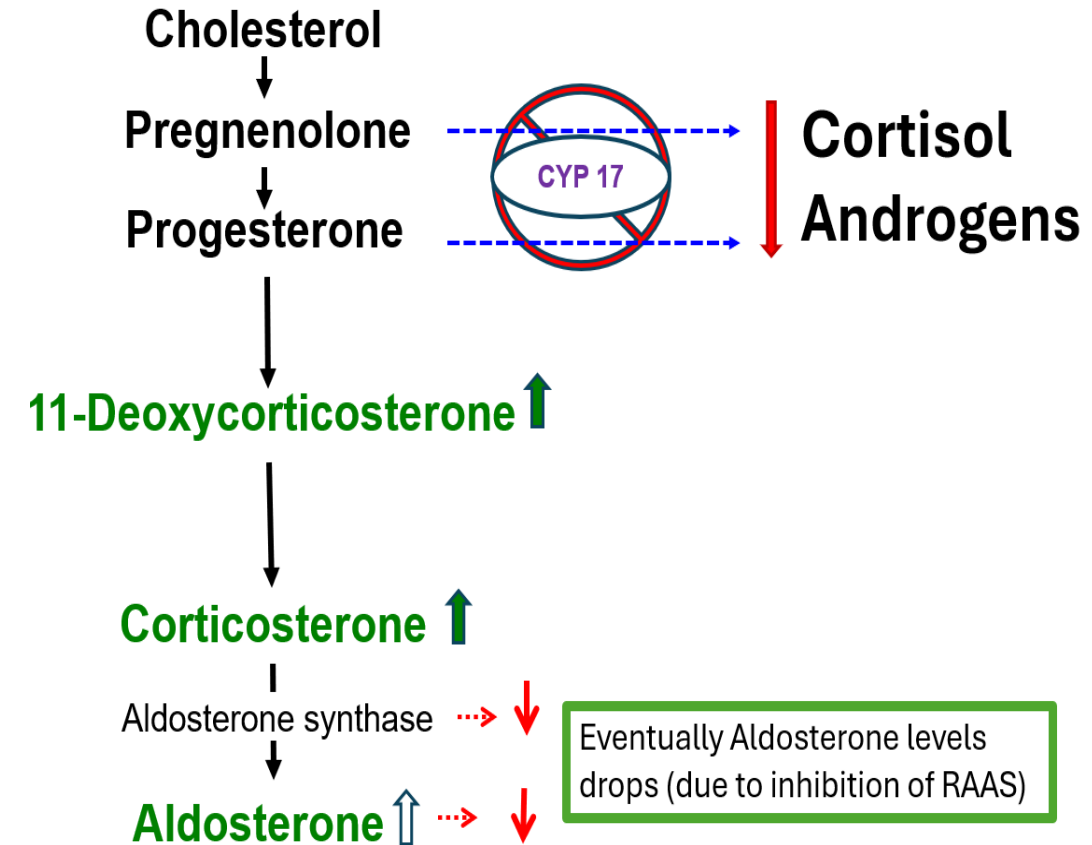
17-hydroxylase deficiency (CYP17)

- **Increased aldosterone and 11-deoxycorticosterone:** Hypertension, Hypernatremia, Hypokalemia
- **Low cortisol:** low blood glucose
- **Low adrenal androgens (DHEA and Androstenedione):**
 - Female newborn: female external genitalia; puberty delay
 - **Male newborn:** ambiguous female-like external genitalia



CAH: Deficiency of CYP 17 (17 α -Hydroxylase) (Feedback Inhibition)

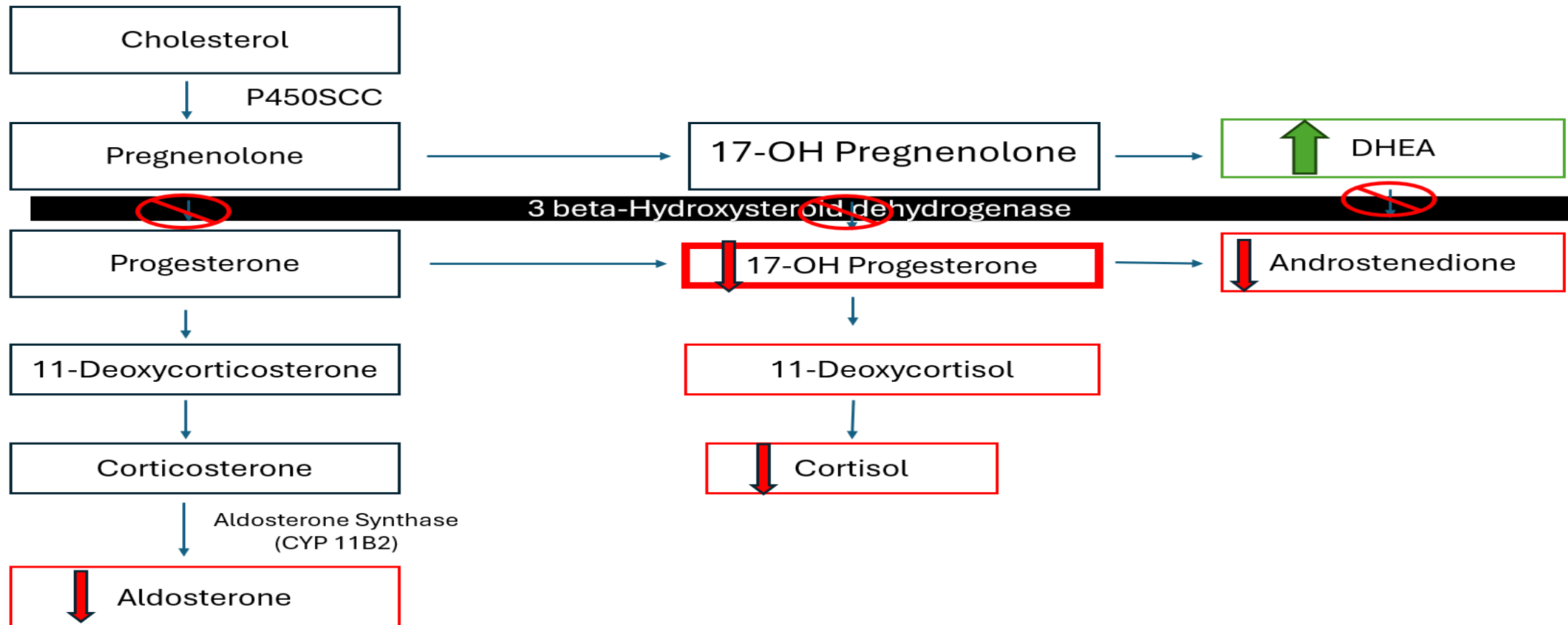
- Initial hypertension: $\uparrow$ ACTH $\rightarrow$ $\uparrow$ 11-deoxycorticosterone, $\uparrow$ corticosterone and $\uparrow$ aldosterone (measured in the blood)
- Overtime: Hypertension $\rightarrow$ inhibits RAAS $\rightarrow$ $\downarrow$ renin $\rightarrow$ $\downarrow$ angiotensin II $\rightarrow$ $\downarrow$ gene expression of aldosterone synthase $\rightarrow$ $\downarrow$ aldosterone synthesis
- Hypertension is due to elevated **11-deoxycorticosterone**



CAH: 3β -hydroxysteroid DH deficiency

Rare

(similar clinical signs like CYP 21 deficiency)



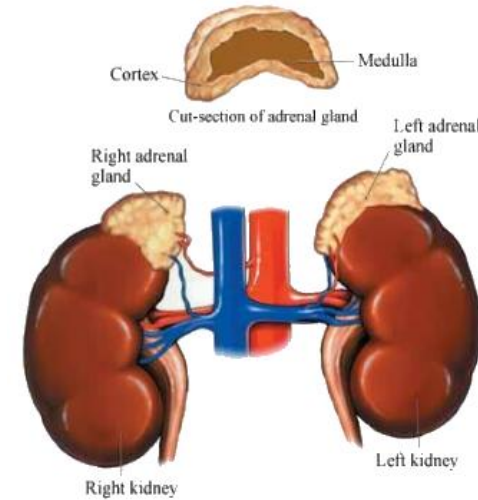
3-beta hydroxysteroid dehydrogenase deficiency

- **Low** aldosterone: Hypotension, Hyponatremia, Hyperkalemia
- **Low** cortisol: low blood glucose
- **Low** Androstenedione
- **Increased** DHEA
 - Female newborn: female or ambiguous male-like external genitalia
 - Male newborn: male or ambiguous female-like external genitalia
- Newborn screening test for 17-OH Progesterone: **NEGATIVE**

De-novo synthesis of steroid hormones in extra adrenal tissues

Adrenal Cortex:

- a. Glucocorticoids: Cortisol
- b. Mineralocorticoids: Aldosterone
- c. Weak Androgens: DHEA and androstenedione

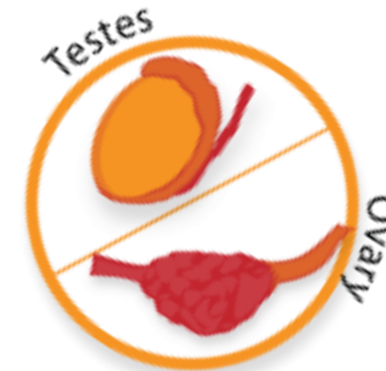


Testes: Androgens: Testosterone

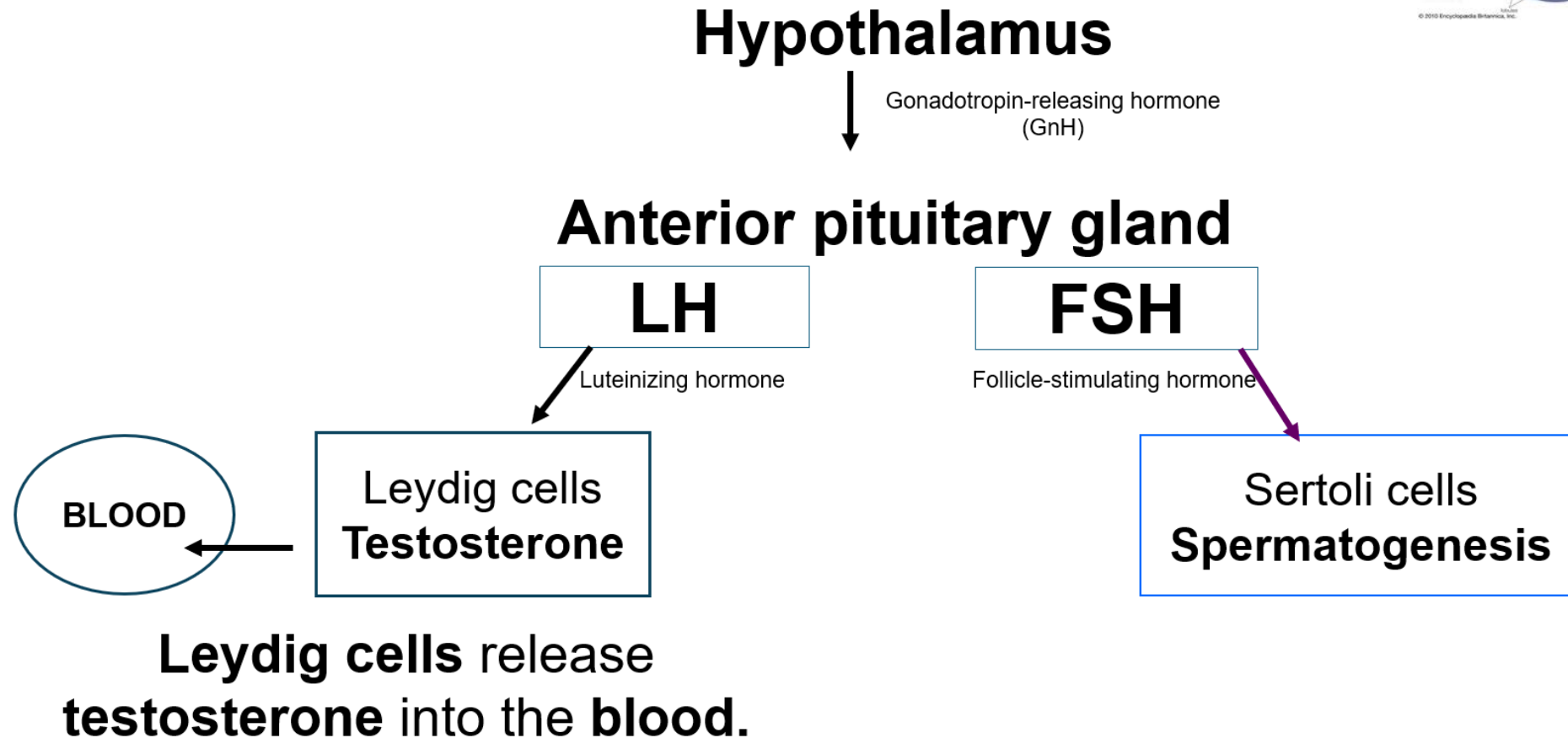
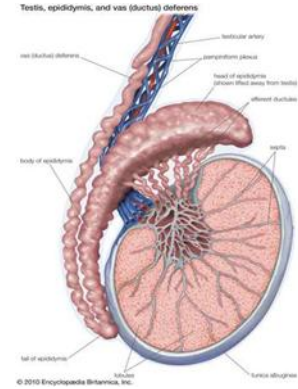


Ovaries: Estrogens: Estradiol

Corpus luteum: Progestins: Progesterone

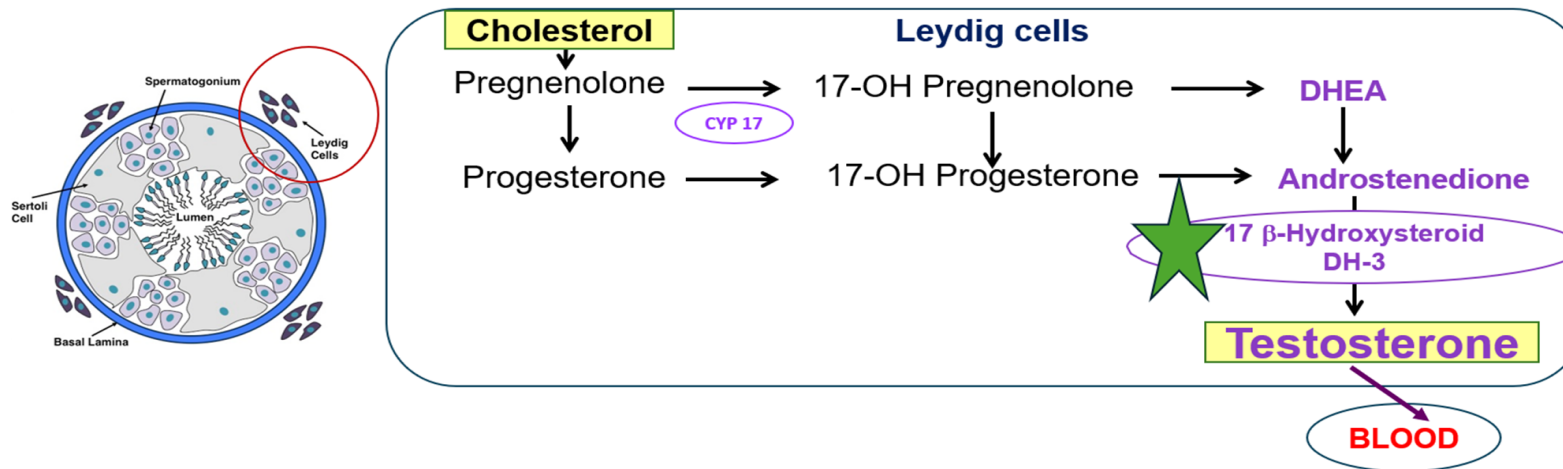


Sex hormone synthesis in testes



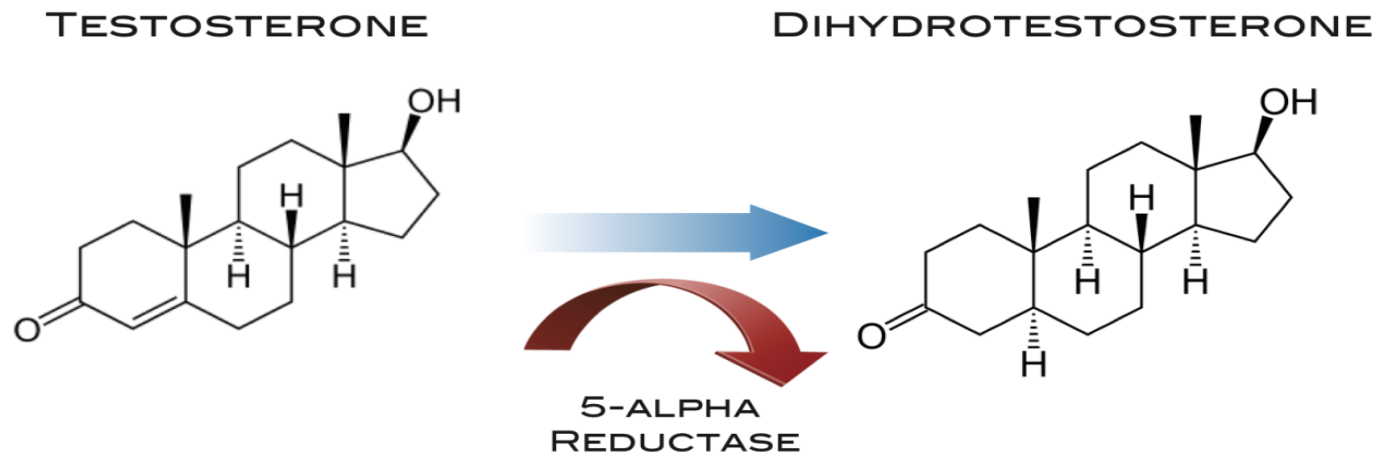
Testosterone synthesis in Leydig cells

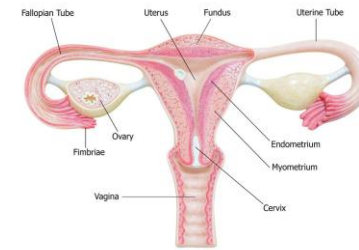
- **Leydig cells**
- Contain 17 β -hydroxysteroid DH-3
- **LH** activates **cAMP** messenger system
- Activation of **testosterone synthesis**
- **Similar pathway to form DHEA and androstenedione as in the adrenal cortex**



Formation of dihydrotestosterone (DHT) by 5-alpha-reductase

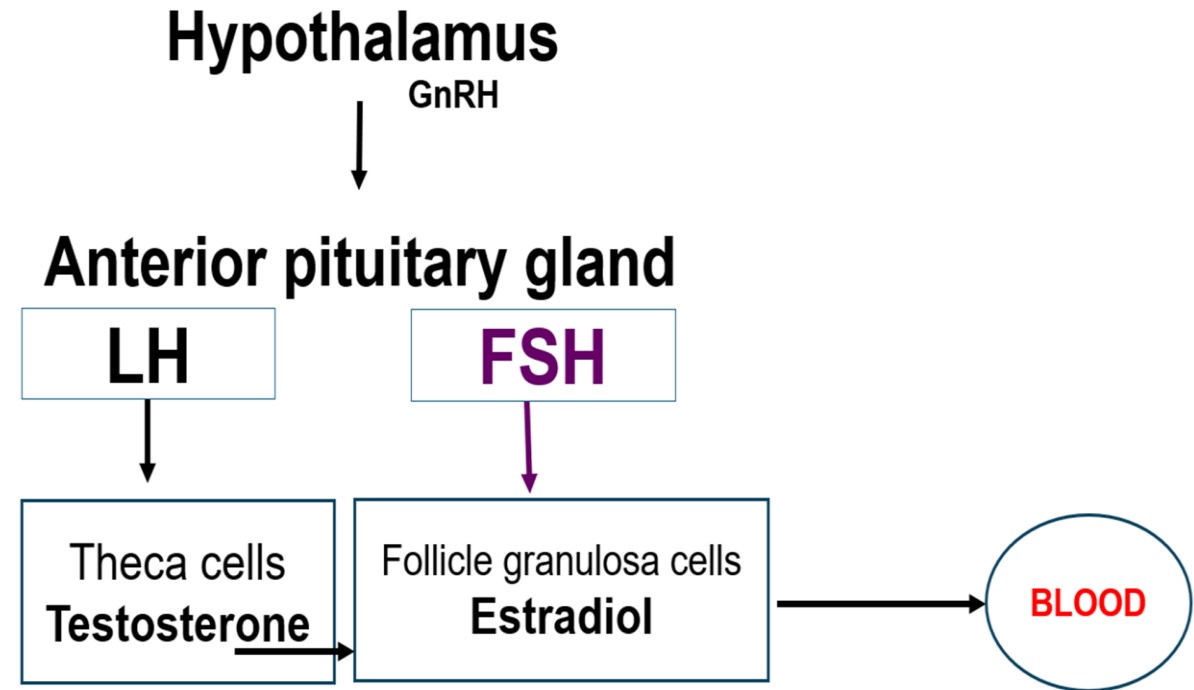
- **Testosterone** released by **Leydig cells**
- **Activated** to **dihydrotestosterone** in **prostate, hair follicles** and other cells.
- **DHT: higher affinity** for the **androgen receptor**.





Sex hormone synthesis in ovaries

- **Follicle granulosa cells**
 - take up **testosterone** from **Theca cells**
 - use it for synthesis of **estradiol**
 - Estradiol released into the blood.



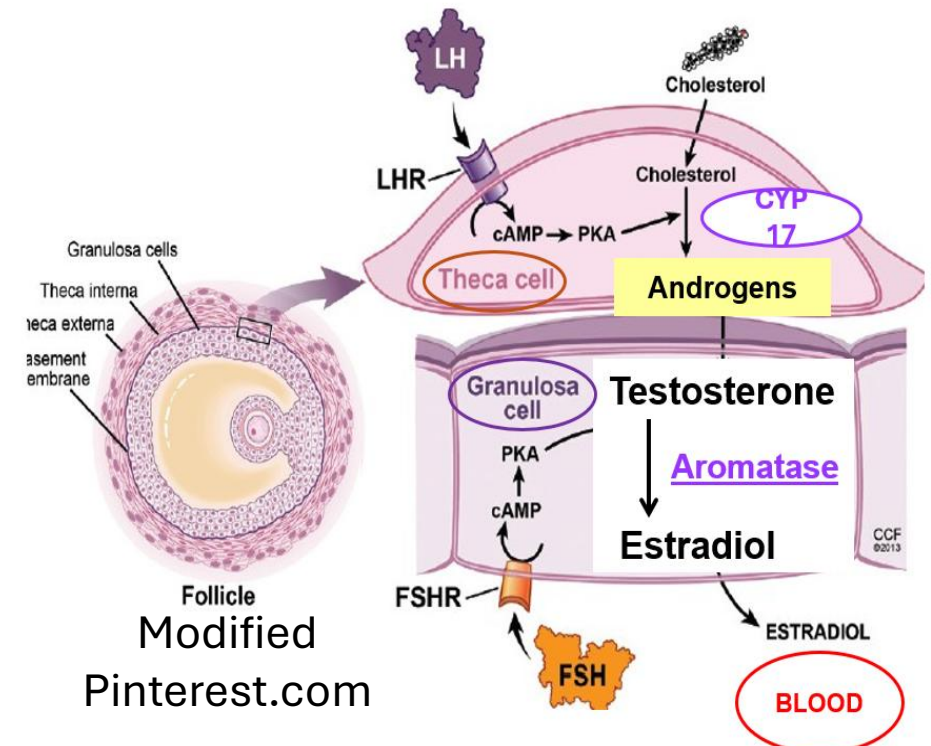
Conversion of androgens to estrogens (Ovaries)

1. Theca cells

- LH
- **Synthesis and secretion of androgens (testosterone and androstenedione)**

2. Follicle granulosa cells

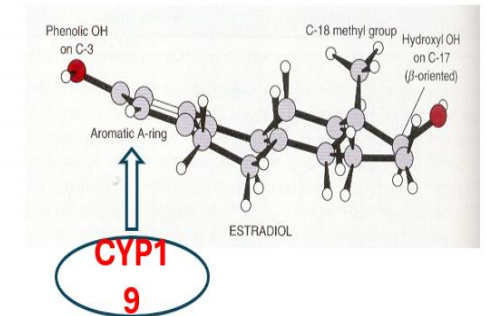
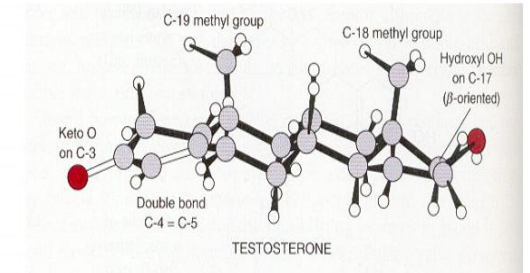
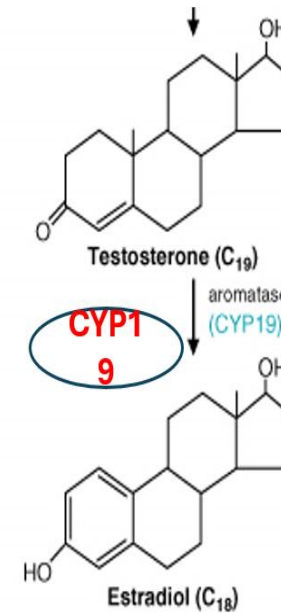
- FSH
- Contain **aromatase (CYP 19)**
- **Use androgens for estrogen synthesis**
- **Release primarily estradiol** formed from **testosterone**



Pinterest.com

Synthesis of estrogens

- **Aromatase (CYP 19)**
- Uses androgens for estrogen synthesis.
- CYP 19 cuts carbon #19
- Forms estrogens (18C) and an **aromatic ring A**
- Aromatic ring **bends** part of the **molecule** upward
- Needed for the **recognition** by the **estrogen receptor**.



Acquired Adrenal Disorders

- 1. Addison Disease: Atrophy of the adrenal cortex**
- 2. Cushing syndrome: Hypercortisolism**

Addison Disease (Primary adrenal insufficiency)

- Insufficiency and atrophy of the adrenal cortex due to:
 - a. Autoimmune destruction or tumors.
 - b. Infections like HIV, tuberculosis, fungi.
- Low level of cortisol lead to:
 - a. Low blood glucose during prolonged gluconeogenesis.
 - b. High level of ACTH does **not** lead to CAH because of the atrophy of the adrenal cortex.
- Low level of aldosterone leads to:
 - a. Low blood pressure and salt craving.
 - b. Low sodium and high potassium blood levels.

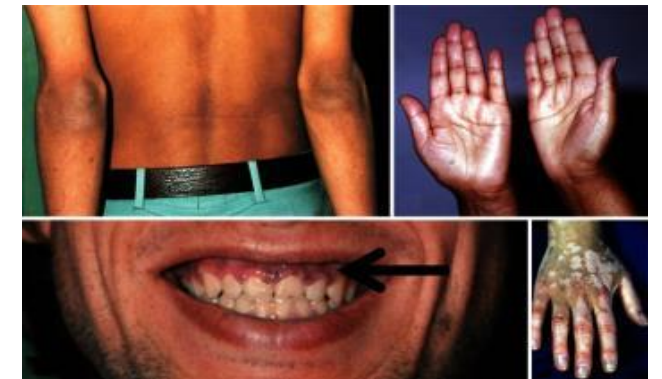
Signs and symptoms of Addison Disease



- **Bronze pigmentation of skin and gum line**
 - high level of **melanocyte stimulating hormone**.
 - Low **cortisol** → increased **CRH** release → activates **proopiomelanocortin (POMC)** cleavage → **ACTH** and **melanocyte stimulating hormone (MSH)**

Note: you find this also in Cushing **disease** (due to high ACTH)

- Failure to thrive
- Weight loss
- Muscle weakness and salt craving
- Thinning of axillary and pubic hair in some females



Cushing Syndrome: Hypercortisolism

- A collection of signs and symptoms due to abnormally high levels of cortisol in the blood resulting from:
 1. Tumor in the Z. fasciculata
 - High cortisol
 - High adrenal androgens
 - Low ACTH (recall: high cortisol inhibits ACTH release)
 2. Tumor in Pituitary (*Cushing Disease*)
 - High cortisol
 - High ACTH
 3. Prolonged administration of high doses of corticosteroidal drugs

Signs and Symptoms of Cushing Syndrome

1. Characteristic fat distribution:

- a. Increased fat in **upper body**.
- b. **Fatty hump** between the shoulders.
- c. **Red round face**.



2. Rapid weight gain with **central obesity**.

3. Pink purple **stretch marks** in the skin (**collagen**).

4. **Hirsutism** and early pubic hair development.



5. **Thin arms and legs, severe muscle weakness** (cortisol stimulates **muscle protein degradation**).

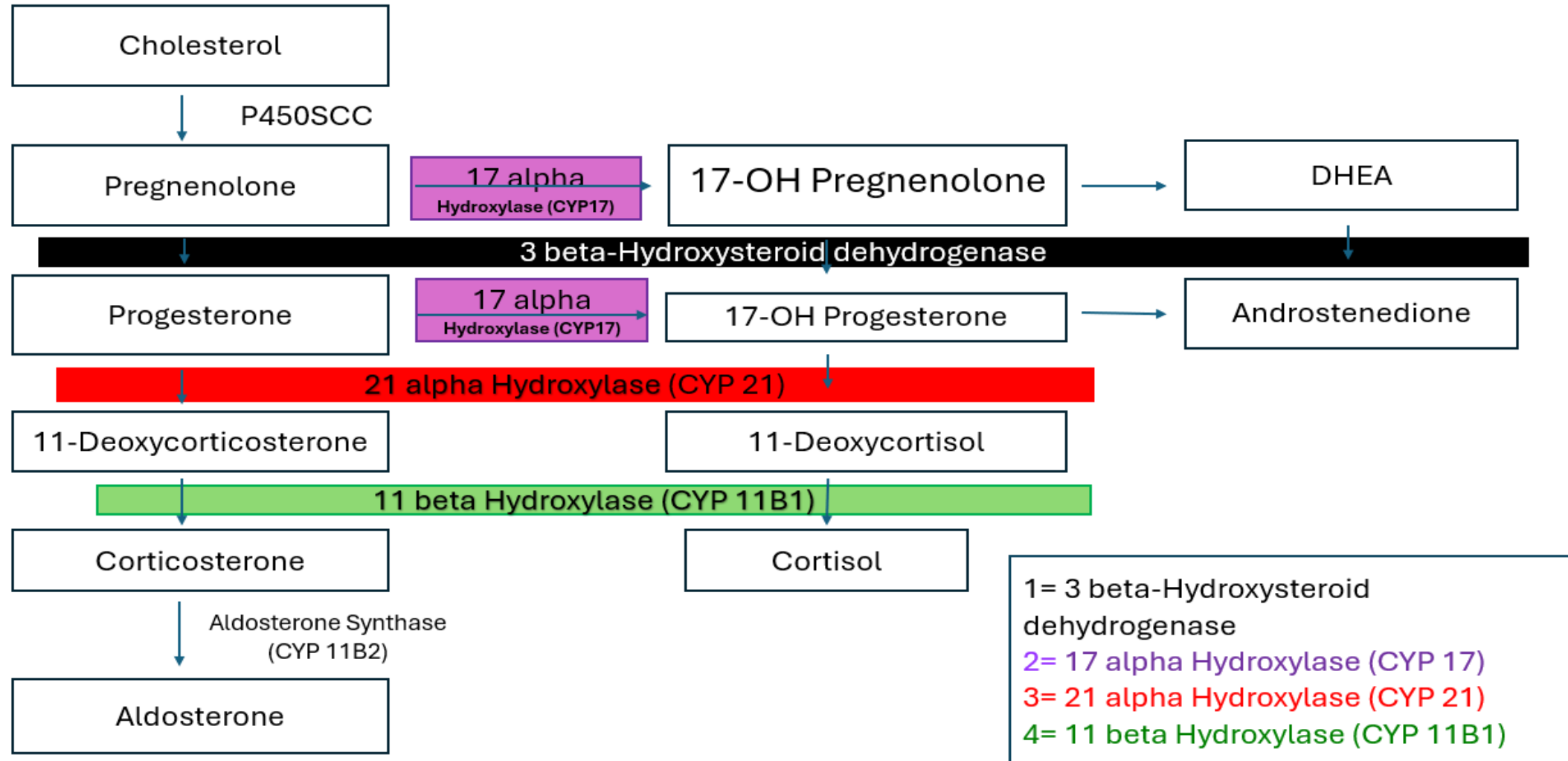
6. **Elevated blood glucose** (cortisol stimulates **gluconeogenesis**).

7. **High blood pressure** from epinephrine or a **mineralocorticoid effect** of cortisol.

8. **Reduced inflammatory response** (cortisol has **anti-inflammatory action**).

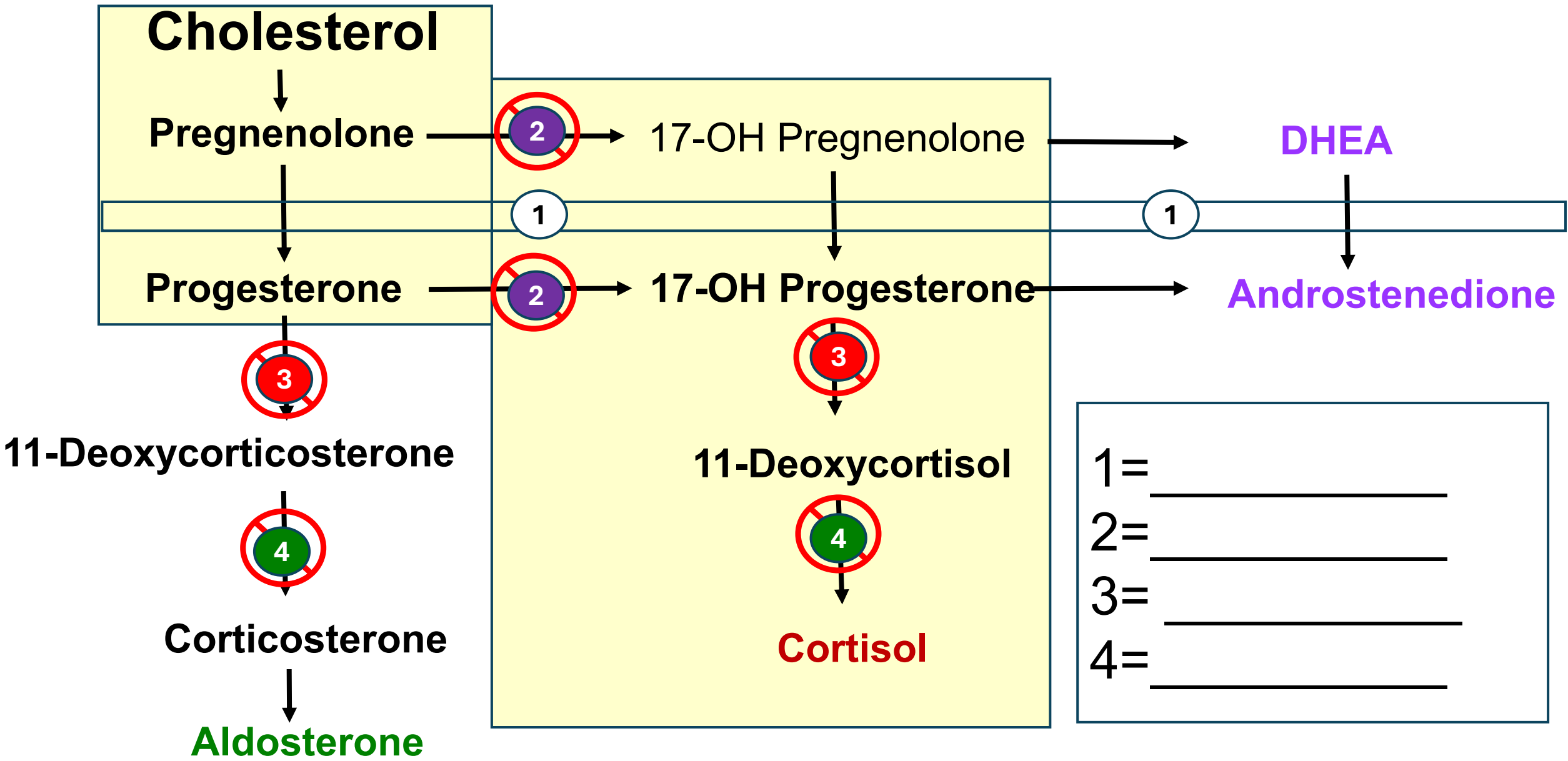
9. **Suppressed Immune System** (reduced protein synthesis in **lymphoid** tissue)

Summary: Enzymes for steroid hormone synthesis in adrenal cortex



Summary	Abnormalities of steroid hormone synthesis	Blood glucose	Blood pressure	Androgens
Hyper-cortisolism: Cushing	Cushing Syndrome: signs and symptoms of abnormally high levels of cortisol. - Tumor in <u>Z. fasciculata</u> (low ACTH) Cushing Disease with <u>pituitary</u> tumor (high ACTH) - Prolonged treatment with corticosteroid drugs S/S: Abnormal fat distribution, face, neck, central obesity, stretch marks - Thin arms and legs (muscle weakness). Increased cortisol & adrenal androgens.	High: Stimulated Gluconeogenesis	Normal to high via cortisol and epinephrine	<u>Increased</u> synthesis in <u>adrenals</u>
Addison Disease	Destruction of adrenals S/S: Hyperpigmentation, bronze gums, weight loss, muscle weakness; Hypoglycemia; hypotension - Females: loss of axillary & pubic hair - Males: mostly normal blood androgen levels Decreased synthesis of all adrenal steroids	Low fasting	Low	<u>Low to absent</u> synthesis in <u>adrenals</u>
CAH: 3 Hydroxy-steroid DH deficiency	All CAH : LOW cortisol Severely decreased cortisol & aldosterone. Androgens are variable - stimulated DHEA in adrenals but reduced sex hormone synthesis in testes and ovaries (Rare)	Low fasting	Low	<u>High DHEA</u> in <u>adrenals</u>
CYP 17 deficiency	Decreased cortisol & general sex hormone synthesis Increased mineralocorticoids - First increase of aldosterone, then 11-deoxycorticosterone and corticosterone - These weaker intermediates add up and lead to hypertension - In some patients, aldosterone levels may be low due to feedback inhibition - Explanation: Hypertension → inhibits RAAS → ↓ renin → ↓ angiotensin II → ↓ gene expression of aldosterone synthase → ↓ aldosterone synthesis	Can be low	High	Low in all: adrenals, testes, ovaries
CYP 21 deficiency	Decreased cortisol & aldosterone Increased adrenal androgens - Severe form: “salt-wasting crisis” Diagnostic test: 17-OH progesterone	Can be low	Low	Increased in adrenals: Elevated DHEA and androstenedione
CYP 11B deficiency	Decreased cortisol - increased 11deoxycorticosterone increased adrenal androgens	Can be low	Moderate high	<u>Increased</u> in <u>adrenals</u> : Elevated DHEA and androstenedione

Worksheet: Name the enzymes and describe the CAH patients



Nomenclature for steroidogenic enzymes

(not meant to memorize but helpful for clarification)

Common Name	Previous Form	Current Form	Gene
Cholesterol side-chain cleavage enzyme, desmolase	P 450 _{scc}	CYP 11A1	<i>CYP11A1</i>
3β-Hydroxysteroid dehydrogenase	3b-HSD	3b-HSD II	<i>HSD3B2</i>
17α-Hydroxylase has 2 activities (17α-hydroxylase and 17,20-lyase)	P450 _{C17}	CYP 17	<i>CYP17</i>
21α-Hydroxylase	P450 _{C21}	CYP 21A2	<i>CYP21A2</i>
11β-Hydroxylase	P450 _{C11}	CYP 11B1	<i>CYP11B1</i>
Aldosterone synthase	P450 _{C11AS}	CYP 11B2	<i>CYP11B2</i>

Thank you!

Please email if you have questions.

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